S-Log MetaRaw

Sony metadata and development controls for DaVinci Resolve 21 · guide · v3.0.0

1 What it is, and what it is not

Sony cameras write the shooting setup into every file: Kelvin, tint, EI, lens, aperture, shutter, colour profile. Resolve uses that data only for FX6 and FX9 MXF files; for MP4 (FX30, FX3, a7 and a6000 series) it ignores it. **S-Log MetaRaw reads it and uses it**: exposure and white balance start from the as-shot values and work in linear light, with the curves and gamuts Sony publishes.

It is not raw. Colour science applied rigorously makes the image behave in a way that reminds you of a raw file: white balance shifts cleanly, exposure moves like a stop of light. But a log MP4 is already compressed, 8 or 10 bit, with the camera's noise reduction applied. On heavy work colour science alone is not enough: as soon as you push past the camera's limits, the missing information in the image shows (banding, noise in lifted shadows, clipped highlights that stay clipped). Exposing well at the camera is still what matters most.

2 The three tools

Script

Workspace › Scripts › S-Log MetaRaw

Reads the metadata of **every clip in the project** and writes it into the Media Pool: Metadata panel, columns, keywords for smart bins, data burn-in, CSV. Fixes the clips' Data Level. *Page 2.*

S-Log MetaRaw node

Color › OpenFX · first node

Develops one clip from its as-shot values: white balance, exposure, colour space, **tones by zones** and false colour. Pointwise: Generate LUT exports it. *Pages 2-3.*

S-Log MetaRaw Detail node

Color › OpenFX · right after

The creative node: highlight and shadow recovery **by areas**, Texture, Clarity, Dehaze. Spatial: Generate LUT leaves it out. *Page 3.*

3 Install

1. Double-click **Installa S-Log MetaRaw.pkg**. It is not signed with an Apple certificate: the first time, right-click › **Open**. It asks for the Mac password (the plugin goes in a system folder).
2. The installer deletes Resolve's plugin cache, so the new panel and the Detail node show up.
3. **Restart DaVinci Resolve.** Every install starts clean. To remove every version: **Disinstalla S-Log MetaRaw.command** (right-click › Open; your CSV exports stay unless you ask).

Nodes	`/Library/OFX/Plugins/SLogMetaRaw.ofx.bundle`
Script	Resolve's Fusion › Scripts › Utility
Library	`/Library/Application Support/SLogMetaRaw`
Needs	macOS 12+, Apple silicon or Intel, DaVinci Resolve 21
Clips	Sony XAVC `.MP4` / `.MXF`; the nodes develop S-Log3, S-Log2, S-Log

4 Workflow

1. **Import** the footage and open the **script**: **1 · Read metadata**, then **2 · Write to Resolve**.
2. On the **Color** page put **S-Log MetaRaw** first: it takes the clip's EI, Kelvin and tint, and at those values it changes nothing. Copied onto another clip, it resets itself with that clip's data.
3. Correct **white balance and exposure** in the node, with the false colour views. Then the **Toni** (tones).
4. For local recovery, Texture, Clarity or Dehaze, add **S-Log MetaRaw Detail** right after.
5. Then the rest of the grade and, **last**, the output CST, LUT or DRT. Until there, stay in log or DWG (page 3).

```
S-Log MetaRaw  →  S-Log MetaRaw Detail  →  rest of the grade  →  output CST / LUT / DRT
```

5 Good to know

- Original files are **never modified**: no transcoding, no rewrapping.
- With non-log profiles (Cine, HLG, S-Cinetone) the nodes stay neutral and say so. For ProRes from external recorders there is *Spazio colore › Senza metadata › Sblocca controlli* (unlock without metadata).
- False colour views **replace the image**: turn them off before you render.
- The version sits at the top of the nodes: once a day it asks GitHub whether there is a newer one (the request carries only the version number) and turns 🟢 if so; a click opens the download in the browser. To turn it off, create the file `~/Library/Application Support/SLogMetaRaw/no_update_check`.
- **Tested only on DaVinci Resolve Studio 21.1 on macOS.** An independent hobby project, distributed as is, with no warranty and no liability for professional use. Open code, GNU GPL v3 licence.

The script Workspace › Scripts › S-Log MetaRaw

One row of buttons, one row of options, the clip list and, below it, everything read from the selected clip, grouped as in Catalyst Browse. The script reads the files; it never modifies them. Its window follows Resolve's language; the nodes' panels are in Italian, so their labels appear here with a translation.

Clips	*Whole Media Pool*, or *Selected clips in the Media Pool*.
1 · Read metadata	One row per clip: camera, lens, aperture, shutter, EI, WB, colour space, data level. The **Status** column says *read*, what changed during the take (*changed: aperture, focus*) or why a clip was skipped. Click a row to see everything.
2 · Write to Resolve	Fills the Media Pool fields: Metadata, columns, keywords, *Camera Notes*, data burn-in. Fixes values Resolve reads wrong from MXF (e.g. *Camera Aperture* F53343 on the FX6).
Export CSV	The values Resolve has no field for (EI, tint, WB mode, focus distance, capture gamma…). In Resolve: *File › Import › Metadata*, with *create custom fields* on.
Camera tag	off · adds camera, gamma and primaries to the keywords.
Overwrite metadata	on · replaces values Resolve already wrote; off, it fills only the empty fields.
Correct the Data Level	on · sets each clip's *Data Level* to Full (log) or Video (709, Cine, HLG). This is the real fix: it holds for the whole project, scopes and exports included.
Also set Input Color Space	off · for colour-managed projects. Careful: a script cannot set it back to *Project*, only you can, by hand.
Version	Bottom right: click it to check GitHub; it turns green when there is a newer release.

> It reads at most 24 samples of the metadata track per clip, within one second: nothing is decoded, and a multi-GB file costs about 100 KB of reading. A disk that does not answer is skipped once, with a message (10 s for the first clip of a volume, 6 s for the others). Running it right after import means the nodes find the data ready.

The S-Log MetaRaw node Color › OpenFX · first node, before CST and LUT

It shows the camera at the top and sets Temperatura, Tint and Esposizione to the as-shot values: there it **changes nothing**. It is a starting point, not a look. Settings are saved per clip; **Rileggi metadata** (re-read) resets everything to the camera values and writes that clip's metadata into the Media Pool.

Ripresa · Decodifica	The shot group. *Clip* lets you change the controls; *Camera metadata* locks them to the as-shot values.
Bilanciamento	*As shot* or presets (Daylight, Cloudy, Shade, Tungsten, Fluorescent, Flash); moving a slider switches it to *Custom*.
Temperatura · Tint	Bradford chromatic adaptation in linear light, from the white the camera recorded. Higher Temperatura warms; positive Tint goes towards magenta.
Punto bianco	Resolve's eyedropper: pick a point that should be neutral and Temperatura and Tint set themselves, with the same maths as the render.
Esposizione (EI)	Double the EI is exactly +1 stop, in linear light, before any curve.
Spazio colore	**Ingresso**: input Spazio colore and Gamma (Automatico asks Resolve), Rilevato, Data level and Livello, with *Ripristina* buttons. **Senza metadata**: see page 3. **Uscita**: output Spazio colore and Gamma, like a Color Space Transform; *Timeline* does not convert. To grade in DWG: *DaVinci WG · DaVinci Intermediate*. Also LUT size and *Esporta LUT (.cube)*.

False colour: one view per control, above its slider

Exposure	Bands in stops around 18% grey, ARRI style: middle grey · one stop over (skin) · near clip · clip · deep shadow · black. The rest in grey.
Temperature	As in CineMatch: the picture turns grey, blue = cool cast, raise Temperatura; orange = warm, lower it. Near-neutral casts are boosted up to 8×.
Tint	green = raise Tint; magenta = lower it. Each view reacts only to its own slider.

> Pick a surface that should be neutral and move the slider **until it stays grey**; alternate Temperatura and Tint a couple of times. For exposure: middle grey in green, skin in pink. The view replaces the image: turn it off before you render.

The S-Log MetaRaw node Toni (tones), Zone, Senza metadata

Tones shows Highlights, Bianco, Soft Clip and Saturation; *Mostra tutti i controlli* adds the rest and the Zone (they also appear by themselves when the node uses them). Highlights is a **film shoulder**; the others are **exposures for a zone**, in stops from 18% grey, where the image moves as with an exposure and texture is kept. No combination can solarise.

Contrast	Turns around the Pivot; at +100 the slope doubles, at −100 it halves, with bounded ends.
Highlights	Negative: compresses the highlights with a slope that falls towards the top, like film; at −100 the recorded maximum lands on **Bianco**, with no grey veil and no clip. Grey and below stay put. Positive: more snap.
Bianco (stop)	Where that maximum lands, and Soft Clip's roof. 2.5 = Rec.709 white through a CST without tone mapping; with a DRT (ACES, AgX, DaVinci) raise it to 4-5.
Shadows · Whites	Shadows: the tones below −1 stop, 100 = 2 stops; lifts black too, hold it with Blacks. Whites: from +3.5 stops to clip, 100 = 1 stop.
Blacks	A linear veil: moves black (−3 / +1 stop) without moving grey.
Vibrance · Saturation	Around luminance, the same in every colour space. Vibrance protects skin tones.
Zone (closed)	Black, Shadow, Light, Specular: Exp (stops), Sat, Range (edge), Falloff (transition). **Soft Clip** folds the highlights under a roof: containment, not recovery. **Zone false colour** shows which zone moves each pixel.
Senza metadata	**Sblocca controlli**: for ProRes or unreadable clips, type the shooting EI, Kelvin and tint and the controls come alive.

Recipes · burning highlights: Highlights −50…−100 (with a DRT after, Bianco at 4-5) · backlight: Shadows +30…+60 and Blacks −20…−40 · film look: Contrast +20…+30, Highlights −60 · noisy shadows: Shadow Sat −30…−50.
The limit, plainly: a pointwise node that compresses a band also compresses its texture. Recovery that keeps it lives in the Detail node. Generate LUT exports this node: 65 points is best.

The S-Log MetaRaw Detail node Color › OpenFX · after S-Log MetaRaw, before the output CST

The creative node: three darkroom methods, each in three **phases** (*Fase*), one node per phase in series or all in one. The **Guida** at the bottom of the panel gives each phase's purpose, three steps and the next step. **Destinazione (ODT)** declares the transform after the node (it never converts): it sets where zone 0 (black) and zone X (white) land. **Misura automatica** reads the frame (2nd/98th percentile) in every mode. Input: scene log only (S-Log3, S-Log2, DaVinci WG/Intermediate, ACEScct), written back in the same encoding.

Zonale	Ansel Adams. Put *Ombra* and *Luce* on the viewer, **Misura zone**, then *Ombra sulla zona* (exposure, III) and *Luci sulla zona* (development N±, VIII). Phases: Sviluppo (per-pixel curve, no halos) · Stampa (local tones, **Brucia i bordi**) · Presenza.
Split grade	Two-exposure print: **Luci morbide** (grade 0) puts the highlights on VIII, **Ombre dure** (grade 5) the shadows on III; **Unione** Curva or Due strati. **Provino**: a test strip on the image; pick a cell and **Usa la cella**.
Unito	Split grade + automatic zones + BTZS: development (N±) brings the subject towards the paper scale (zones III–VIII), the print does the rest.
Recommended	Main node on *DaVinci WG* · *DaVinci Intermediate* → Detail with *Ingresso (IDT)* on *DaVinci WG/Intermediate* → grade in DWG → last, an **output CST** DaVinci WG/Intermediate → Rec.709 · Gamma 2.4, DaVinci tone mapping. With tone mapping after, set **Bianco** to 4-5 in both nodes.
All in log	Main node on Timeline → Detail on Automatic (uses the clip's S-Log3) → CST S-Gamut3.Cine/S-Log3 → Rec.709 or DWG.
Gamma dinamica	dynamic range · *Vista: zone* (colours or greys, 0–X legend) · **Local Highlights** compresses the large bright areas and keeps, even strengthens, the texture: the sky gets deeper and the clouds stay defined. **Local Shadows** opens (+) or closes (−) shadows; **Local Contrast** the contrast of large areas. *Gain* and *base* views.
Presenza	presence · **Texture**: fine detail; does not raise grain below the noise threshold; negative smooths. **Clarity**: local contrast on the midtones. **Dehaze**: removes (+) or adds (−) haze.
Avanzate · Velo	advanced · haze · detail preservation, radius, **Soglia bordi** (edge threshold), noise threshold, Clarity centre, **Bianco** for Local Highlights. You set the haze (level and colour): never guessed per frame.

Measured limits · halo under 3% with Local Highlights −100, up to 12% on a hard one-stop edge with Local Shadows ±100: lower *Soglia bordi*. Grain can grow 1.25-1.7× next to strong edges. It is **spatial**: Generate LUT leaves out its whole node. On Metal a UHD frame costs about 6-18 ms.